CLOUD COMPUTING PAST PAPER ANSWERS

SEPTEMBER 2018

QUESTION 1

a) Define cloud computing (2 marks)

Cloud computing refers to the delivery of computing services—such as servers, storage, databases, networking, software, and analytics—over the internet ("the cloud") to offer faster innovation, flexible resources, and economies of scale.

b) Outline six characteristics of cloud computing (6 marks)

On-Demand Self-Service: Users can access computing resources automatically without human intervention.

Broad Network Access: Services are accessible over the internet through various devices like phones, laptops, and desktops.

Resource Pooling: Resources are pooled to serve multiple users dynamically, enabling resource sharing and multi-tenancy.

Rapid Elasticity: Resources can scale up or down quickly to meet demand.

Measured Service: Usage is monitored and billed on a pay-as-you-go basis.

Multi-Tenancy: Multiple users share the same physical resources securely.

c) Outline three types of clouds (6 marks)

Public Cloud: Services are provided over the internet to multiple users by third-party providers (e.g., AWS, Azure).

Private Cloud: Dedicated resources for a single organization, ensuring higher security and control.

Hybrid Cloud: Combines public and private clouds, allowing data and applications to be shared between them.

d) Define virtualization (2 marks)

Virtualization is the process of creating virtual versions of physical resources, such as servers, storage, or networks, allowing multiple operating systems or applications to run simultaneously on the same hardware.

e) Discuss three cloud service models (6 marks)

Infrastructure as a Service (IaaS): Provides virtualized computing resources (e.g., servers, storage) over the internet. Users manage applications, data, and OS, while the provider manages hardware.

Platform as a Service (PaaS): Offers a platform for developers to build, test, and deploy applications without managing the underlying infrastructure.

Software as a Service (SaaS): Provides software applications over the internet on a subscription basis, eliminating the need for installation or maintenance by the user.

f) Outline six security issues in cloud computing (6 marks)

Data Breaches: Unauthorized access to sensitive data.

Data Loss: Permanent loss of data due to provider failures or malicious attacks.

Insider Threats: Malicious or careless actions by authorized individuals.

Denial of Service (DoS): Attacks that disrupt access to cloud services.

Account Hijacking: Unauthorized access to cloud accounts via phishing or stolen credentials.

Compliance Issues: Failure to meet regulatory requirements for data protection and privacy.

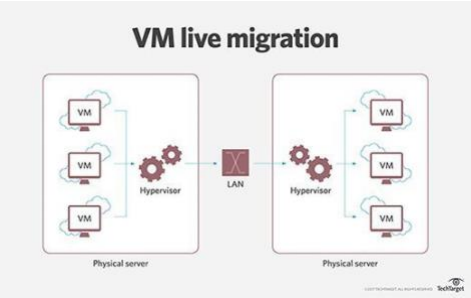
QUESTION 2

a) Using appropriate diagrams, describe the following terminologies as used in business

continuity in cloud computing

i. Virtual machine live migration (4 marks)

Virtual Machine Live Migration refers to the process of moving a running virtual machine (VM) from one physical host to another without interrupting the VM's operations. This is essential for load balancing, hardware maintenance, or disaster recovery.



ii. Link aggregation (4 marks)

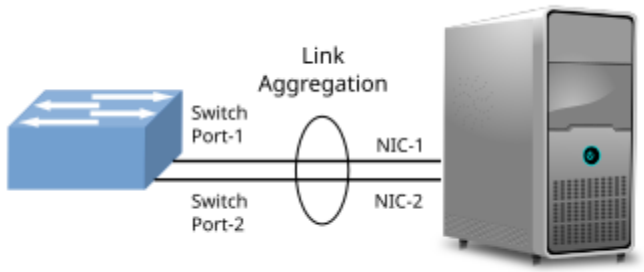
An aggregation link, also known as a link bundle, is a logical connection that combines multiple network connections into one. This technique is used to increase bandwidth, improve performance, and provide redundancy.

How it works

Link aggregation combines multiple network interface cards (NICs) into a single logical interface.

The aggregate link speed is faster than the individual network interface speeds.

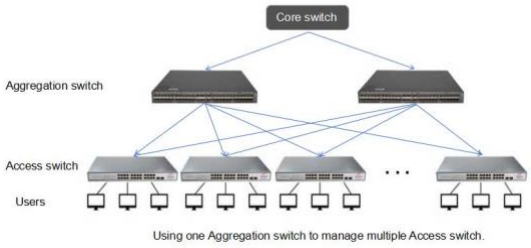
If one connection fails, traffic is automatically directed to the other connection.



Link aggregation between a switch and a server

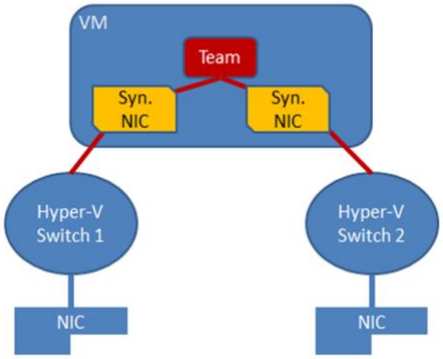
iii. Switch aggregation (4 marks)

Switch Aggregation involves connecting multiple switches together to function as a single logical switch, allowing seamless communication and redundancy between devices.



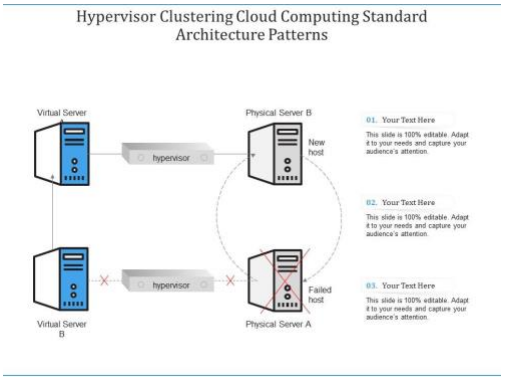
iv. NIC teaming (4 marks)

NIC Teaming is the practice of grouping multiple Network Interface Cards (NICs) on a server to provide redundancy and increase network performance. If one NIC fails, traffic is rerouted to the others.



v. Hypervisor clustering (4 marks)

Hypervisor Clustering refers to the grouping of multiple hypervisors (virtualization platforms) to provide high availability, load balancing, and fault tolerance for virtual machines.



QUESTION 3

Discuss the application of cloud computing solutions in the following areas

a) Energy Systems (5 marks)
Cloud computing solutions in energy systems enable efficient management, monitoring, and optimization of energy production, distribution, and consumption.

1. **Smart Grids**: Cloud computing enables real-time monitoring and management of electricity distribution.
2. **Energy Analytics**: It provides data analytics to predict energy demand and optimize resource usage.
3. **Renewable Energy Management**: Facilitates monitoring and integration of renewable energy sources like solar and wind.
4. **Energy Storage**: Helps optimize battery usage and storage systems using cloud-based algorithms.
5. **IoT Integration**: Supports IoT devices for remote monitoring and control of energy systems.

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b) Transportation Systems (5 marks)
Cloud computing enhances transportation systems by enabling real-time tracking, optimization, and automation.

1. **Fleet Management**: Real-time tracking and scheduling of vehicles for efficiency and cost reduction.
2. **Traffic Management**: Analyzing traffic data to predict and manage congestion.
3. **Public Transportation**: Supports ticketing systems, route optimization, and real-time updates for passengers.

4. **Autonomous Vehicles**: Provides the computational power required for AI and data processing.
5. **Logistics Optimization**: Enables cloud-based route planning and warehouse management.

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c) Manufacturing Industry (5 marks)
Cloud computing transforms manufacturing by enhancing productivity, automation, and cost-efficiency.

1. **Supply Chain Management**: Enables real-time tracking of raw materials and products.
2. **Predictive Maintenance**: Monitors equipment to predict failures and reduce downtime.
3. **Smart Factories**: Supports IoT-enabled manufacturing processes for automation and efficiency.
4. **Product Design and Testing**: Facilitates collaborative design and simulation in virtual environments.
5. **Data Analytics**: Analyzes production data to improve processes and reduce waste.

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d) Healthcare (5 marks)
Cloud computing revolutionizes healthcare by improving patient care, data storage, and operational efficiency.

1. **Electronic Health Records (EHRs)**: Centralized storage and sharing of patient data among healthcare providers.
2. **Telemedicine**: Enables remote consultations and real-time communication between patients and doctors.
3. **Medical Research**: Facilitates data analysis and collaboration for developing new treatments.
4. **Wearable Devices**: Collect and analyze patient data in real time using cloud-based platforms.
5. **Scalable IT Infrastructure**: Supports hospital systems with scalable resources to manage patient data and operations.

<u>QUESTION 4</u>

a) You have been identified as an expert in cloud computing. You are expected to share a talk on software as a service (saas). In your answer, briefly describe 4 saas maturity models. Illustrate your answer with suitable example (8 marks)

1. **Ad Hoc/Custom SaaS**:
   Applications are tailored for a specific client and hosted on a separate environment.
   Example: A company-specific HR management tool hosted by the vendor.
2. **Configurable SaaS**:
   The application is shared across multiple clients (multi-tenancy) but allows extensive customization.
   Example: Salesforce, where businesses can configure workflows and dashboards to fit their needs.
3. **Scalable SaaS**:
   Applications support large-scale usage with optimized architecture and automated scaling for high demand.
   Example: Zoom, which scales video conferencing infrastructure automatically during peak demand.
4. **Fully Mature SaaS**:
   The application has advanced capabilities, such as AI-driven insights, seamless integrations, and full multi-tenancy.
   Example: Google Workspace (formerly G Suite), with integrated tools like Gmail, Drive, and Meet.

b) Using appropriate example, describe 4 security features that devices that access the

cloud should have (8 marks)

- **Endpoint Encryption**:
Encrypts data stored on the device and data in transit to protect sensitive information.
Example: A smartphone using AES-256 encryption for files and communications.

- **Multi-Factor Authentication (MFA)**:
Requires users to verify their identity through multiple methods (e.g., password and biometric verification).
Example: A user logging into a cloud service using a password and fingerprint scan.

- **Device Management Tools**:
Ensures compliance with security policies by remotely managing, wiping, or locking devices.
Example: Microsoft Intune for managing and securing devices accessing Microsoft 365.

- **Firewalls and Antivirus Software**:
Protects devices from malware, unauthorized access, and network attacks.
Example: Devices equipped with tools like Norton or McAfee to block malicious traffic.

c) If you were to advice a cloud service provider in hiring new staff, briefly explain 4

emerging professionals you would recommend. What would be their role? (4 marks)

- **Cloud Security Specialist**:

Role: Designs and implements strategies to secure cloud infrastructures and ensure data protection.

- **DevOps Engineer**:

Role: Manages cloud-based development pipelines and ensures efficient integration and deployment of applications.

- **AI/ML Specialist**:

Role: Develops artificial intelligence and machine learning models to enhance cloud services with analytics and automation.

- **Cloud Solutions Architect**:

Role: Designs comprehensive cloud strategies and solutions tailored to client needs, ensuring scalability and efficiency.

QUESTION 5

a) Briefly explain three roles/functions of hypervisor layer (6 marks)

1. **Virtual Machine Management**:
   The hypervisor creates, manages, and runs multiple virtual machines (VMs) on a single physical host, enabling efficient resource sharing.
   Example: VMware manages multiple virtual machines on a server for different clients.
2. **Hardware Abstraction**:
   It abstracts hardware resources like CPU, memory, and storage, allowing VMs to use them as independent virtual resources.
   Example: A hypervisor allocates portions of CPU and memory to each VM.
3. **Isolation and Security**:
   Ensures that VMs operate in isolated environments, preventing one VM from interfering with others or accessing unauthorized resources.
   Example: Xen hypervisor isolates VMs for enhanced security in cloud environments.

b) Provide four major reasons why cloud services might become unavailable (4 marks)

- **Network Connectivity Issues**:

Loss of internet connectivity or poor bandwidth can disrupt access to cloud services.

- **Data Center Outages**:

Power failures or hardware malfunctions in the cloud provider's data center can lead to service downtime.

- **Cyberattacks**:
Distributed Denial of Service (DDoS) attacks or other malicious activities can overwhelm cloud infrastructure and disrupt services.

- **Software Failures**:
Bugs or misconfigurations in cloud software or platforms can result in service unavailability.

c) Using appropriate examples, briefly describe five payment models commonly used in

cloud computing (10 marks)

- **Pay-As-You-Go**:
Users are charged based on actual usage of cloud resources, such as compute time or storage space.
Example: Amazon Web Services (AWS) charges for the exact number of hours a virtual machine runs.

- **Subscription-Based**:
Users pay a fixed fee for a defined period (e.g., monthly or annually) to access services.
Example: Microsoft 365 subscription for access to cloud-based productivity tools.

- **Free Tier**:
Users get limited access to services for free, often with options to upgrade for additional features.
Example: Google Cloud offers free usage credits for basic storage and compute services.

- **Spot Pricing**:
Users can bid for unused cloud resources at lower prices, ideal for non-critical workloads.
Example: AWS Spot Instances allow bidding on unused EC2 capacity.

- **Resource-Based Pricing**:
Costs are determined by specific resources consumed, such as per GB of storage or per transaction.
Example: Google Cloud charges based on the amount of data stored or queried in BigQuery.

JANUARY 2019

QUESTION 1

a) Using appropriate examples, describe 3 security features that devices access the cloud

should have (6 marks)

- **Multi-Factor Authentication (MFA)**: Ensures only authorized users can access the cloud by requiring multiple forms of identity verification.
Example: A user logging into AWS with a password and a one-time code sent to their phone.

- **Endpoint Encryption**: Protects data stored on and transmitted from the device by encrypting it.
Example: A laptop with BitLocker encryption accessing Google Drive.

- **Firewall and Antivirus Protection**: Blocks unauthorized access and detects malware threats.
Example: A device using Norton antivirus and a firewall to protect cloud-related traffic.

b) Provide 4 major reasons why cloud services might become unavailable (4 marks)

1. **Data Center Outages**: Caused by power failures, natural disasters, or hardware malfunctions.
2. **Network Connectivity Issues**: Poor internet connectivity or network interruptions.
3. **Cyberattacks**: Attacks like DDoS can overwhelm cloud resources and disrupt services.
4. **Misconfigurations**: Errors in cloud system setup or maintenance.

c) Briefly explain 3 roles/functions of hypervisor layer (6 marks)

1. **Resource Allocation**: Distributes physical hardware resources (CPU, RAM, etc.) to multiple virtual machines (VMs).
2. **Isolation**: Ensures VMs operate independently to maintain security and prevent resource conflicts.
3. **VM Management**: Handles the creation, deletion, and maintenance of VMs.

d) You have been identified as an expert in cloud computing. You are expected to share a

talk on software as a service (SaaS). In your answer, describe 2 SaaS maturity models.

Illustrate your answer with suitable examples (4 marks)

1. **Custom SaaS**: Tailored applications for specific customers hosted on dedicated environments.
   Example: A company-specific accounting tool provided by a SaaS vendor.
2. **Scalable SaaS**: Supports multiple clients with shared resources, offering customization and scaling.
   Example: Google Workspace (formerly G Suite) allowing organizations to scale user accounts.

e) What are the two I/O virtualization subsystem architecture? (2 marks)

- **Direct Assignment Architecture**
- **Mediated Pass-Through Architecture**

f) List three emulation implementation techniques (3 marks)

- **Interpretation**
- **Binary Translation**
- **Hardware-Assisted Emulation**

g) What are the two types of virtual machine hypervisors (2 marks)

- **Type 1 (Bare-Metal Hypervisors)**: Runs directly on the hardware (e.g., VMware ESXi).
- **Type 2 (Hosted Hypervisors)**: Runs on top of a host operating system (e.g., VirtualBox).

h) Discuss the three hardware virtualization techniques (3 marks)

- **Full Virtualization**: Uses a hypervisor to provide complete hardware emulation.
- **Paravirtualization**: Requires modifications to the guest OS to optimize performance.
- **Hardware-Assisted Virtualization**: Leverages hardware features like Intel VT-x or AMD-V to support virtualization.

QUESTION 2

a) What is cloud governance? Using appropriate examples, briefly describe 5 reasons

why cloud governance is important (12 marks)
Cloud governance refers to the framework of policies, roles, and processes that ensure the secure, efficient, and compliant use of cloud resources within an organization. It involves monitoring cloud operations, managing risks, and ensuring alignment with business objectives.

Importance of Cloud Governance:

1. **Cost Management and Optimization**:
   Enables organizations to control and optimize cloud expenses by monitoring resource usage and avoiding unnecessary spending.
   Example: A company implementing policies to terminate idle virtual machines automatically.
2. **Security and Risk Management**:
   Ensures that security measures like data encryption, access controls, and compliance standards are enforced.
   Example: Using Identity and Access Management (IAM) policies to restrict sensitive data access.
3. **Regulatory Compliance**:
   Ensures adherence to industry regulations such as GDPR or HIPAA to avoid legal penalties.

Example: Healthcare organizations managing cloud-stored patient data under HIPAA compliance.

4. **Operational Efficiency**:
   Establishes standardized processes to streamline cloud resource provisioning and usage.
   Example: Automating infrastructure deployment through Infrastructure-as-Code (IaC) tools.
5. **Disaster Recovery and Business Continuity**:
   Ensures robust backup and recovery mechanisms for uninterrupted business operations.
   Example: Configuring automated backups and replication across different cloud regions.

b) Briefly explain the following concepts in the context of cloud computing

i. In-service software upgrade (ISSU)entails in the context of cloud computing (4 marks)

ISSU refers to the process of upgrading software (e.g., operating systems, applications, or firmware) in cloud environments without disrupting services. This ensures continuous system availability and minimizes downtime during upgrades.

Example: A cloud provider upgrading the Kubernetes orchestration system while running customer applications seamlessly.

ii. DRaaS (4 marks)

DRaaS is a cloud-based solution that provides backup and recovery capabilities to restore IT infrastructure and operations in case of a disaster. It ensures business continuity by replicating data and systems to a remote cloud environment.

Example: A retail company using AWS Elastic Disaster Recovery to recover applications and data in the event of a server failure or cyberattack.

QUESTION 3

a) Outline the three cloud computing layers (6 marks)

- **Infrastructure as a Service (IaaS)**:
Provides virtualized computing resources such as virtual machines, storage, and networks.
Example: Amazon EC2, Microsoft Azure Virtual Machines.

- **Platform as a Service (PaaS)**:
Provides a platform and environment for developers to build, test, and deploy applications without managing infrastructure.
Example: Google App Engine, Microsoft Azure App Services.

- **Software as a Service (SaaS)**:
Delivers software applications over the internet, accessible via web browsers without

installation.
Example: Google Workspace, Salesforce CRM.

b) Describe the application and infrastructure focused cloud computing layers services (6 marks)

1. **Application-Focused Services (SaaS)**:
   These services provide end-user software applications via the cloud.
   Examples:
   - Google Docs for document editing.
   - Dropbox for cloud storage.
2. **Infrastructure-Focused Services (IaaS)**:
   These services provide the underlying infrastructure, including virtual machines, storage, and networking, required to host applications.
   Examples:
   - Amazon S3 for object storage.
   - Microsoft Azure Virtual Network for networking.

c) Using a diagram differentiate factors for virtual machine on multi-core CPU's (8 marks)

- **CPU Core Allocation**: Each virtual machine (VM) is assigned a specific number of cores from the multi-core CPU.
- **Resource Isolation**: VMs operate independently, preventing one VM from impacting others.
- **Resource Sharing**: Hypervisors allocate shared resources among VMs efficiently.
- **Performance Scaling**: Multi-core CPUs enable better load balancing and parallel execution for VMs.

QUESTION 4

d) Outline a diagram illustrate the two modes of CPU architecture (6 marks)

In CPU architecture, "user mode" refers to a restricted execution environment where user applications run, providing limited access to system resources and prohibiting direct hardware manipulation.

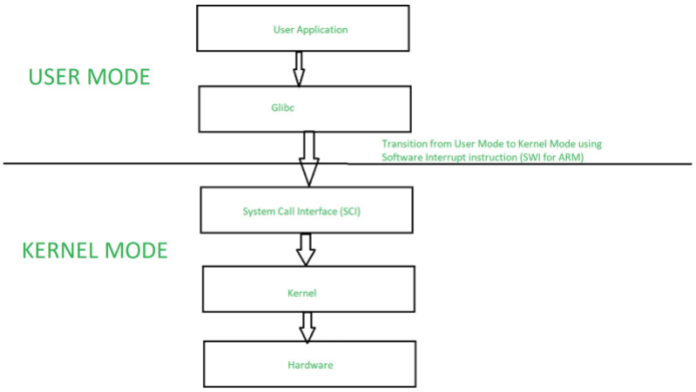
User Mode:

- Limited access to hardware resources.
- Used for executing user applications.
- Transition to kernel mode occurs via system calls.

In a CPU architecture, "kernel mode" refers to a privileged execution state where the operating system kernel runs with unrestricted access to all system resources and hardware.

Kernel Mode:

- Full access to hardware resources.
- Used for executing OS-level processes like hardware management.



e) Highlight three goals of memory virtualization (6 marks)

- **Efficient Memory Utilization**:
Ensures that physical memory is used optimally by distributing resources among multiple virtual machines.

- **Isolation and Security**:
Protects the memory space of one virtual machine from being accessed or corrupted by another.

- **Scalability**:
Enables dynamic allocation of memory resources based on demand, supporting the scalability of applications.

a) Describe the limiting factors for virtual machine on mult-core CPUs (8 marks)

- **Overhead from Hypervisors**:
Hypervisor processes consume CPU cycles, reducing the performance available for virtual machines.

- **Resource Contention**:
Multiple VMs sharing the same CPU cores can lead to contention, especially during high workloads.

- **Non-Uniform Memory Access (NUMA)**:
VMs may experience latency when accessing memory on a multi-core CPU, depending on the memory's physical location.

- **I/O Bottlenecks**:
High I/O operations (e.g., disk or network) may limit CPU performance due to slow peripheral devices.

- **Scalability Limitations**:
As the number of VMs increases, the ability to allocate equal CPU power decreases.

- **Scheduling Delays**:
CPU scheduling by the hypervisor may introduce delays, especially with time-sensitive applications.

- **Core Affinity**:
Poor management of core affinity (assigning specific cores to VMs) can reduce performance.

- **Lack of Optimization for Parallelism**:
Some applications may not fully utilize the parallel processing capabilities of multi-core CPUs.

QUESTION 5

a) Provide 5 major reasons why cloud services might become unavailable (10 marks)

- **Network Connectivity Issues**:
Cloud services depend on internet connectivity. Any disruption in network infrastructure, such as outages, can cause unavailability.
Example: A regional ISP outage preventing access to AWS services.

- **Hardware Failures**:
Physical servers hosting cloud services might fail due to issues like power outages or component damage.
Example: Disk failure in a data center leading to service downtime.

- **Cybersecurity Attacks**:
DDoS attacks or other security breaches can overwhelm cloud servers, causing them to become unresponsive.
Example: A DDoS attack targeting a cloud provider's web servers.

- **Software Bugs or Configuration Errors**:
Faulty updates, software glitches, or misconfigured systems can disrupt cloud service operations.
Example: A misconfigured firewall rule blocking access to critical cloud applications.

- **Natural Disasters**:
Events like earthquakes, floods, or fires affecting data centers can render cloud services unavailable.
Example: A hurricane damaging a data center and disrupting cloud services in the region.

b) Using appropriate examples, briefly describe 5 payment models commonly used in cloud computing (10 marks)

- **Pay-as-You-Go**:
Users pay only for the resources they consume, such as compute power, storage, or bandwidth.
Example: Amazon Web Services (AWS) charges based on hourly usage of virtual machines.

- **Subscription-Based Model**:
Users pay a fixed fee for a specific period (monthly or annually) for access to cloud services.
Example: Microsoft 365 offers annual subscriptions for productivity tools.

- **Resource-Based Pricing**:
Charges are based on the specific resources utilized, such as memory, CPU cycles, or storage space.
Example: Google Cloud Platform pricing for storage per gigabyte used.

- **Tiered Pricing**:
Costs decrease as usage increases, with predefined pricing tiers offering discounts for higher consumption levels.
Example: AWS S3 storage provides reduced rates for higher storage volumes.

- **Freemium Model**:
Basic cloud services are provided for free, with premium features available at a cost.
Example: Dropbox offers free storage for a limited capacity, with paid plans for additional storage and features.

SEPTEMBER 2019

QUESTION 1

a) Briefly describe three types of clouds. (6 marks)

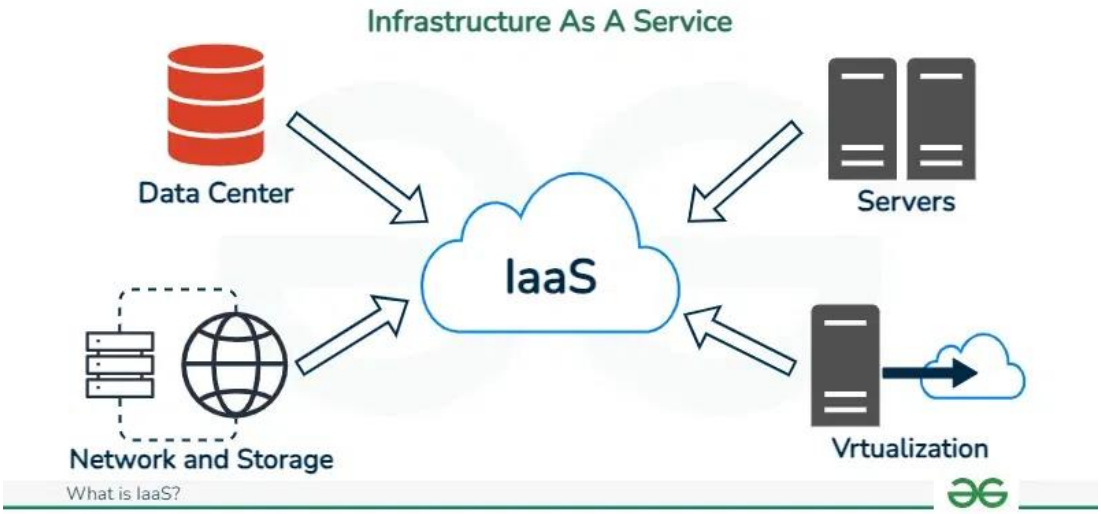
1. **Public Cloud**:
   Resources like servers and storage are owned and operated by third-party providers and delivered over the internet.
   Example: Google Cloud, Amazon Web Services.
2. **Private Cloud**:
   Exclusively used by a single organization. It provides greater control over data and resources.
   Example: An on-premise data center managed by a company.
3. **Hybrid Cloud**:
   Combines public and private clouds, allowing data and applications to be shared between them for flexibility and scalability.
   Example: A company using both Azure Public Cloud and its private cloud infrastructure.

b) Which resources would you need to set up a local cloud? (3 marks)

1. **Hardware Resources**: Servers, storage devices, and networking equipment.
2. **Virtualization Software**: Tools like VMware or Hyper-V for creating virtualized environments.
3. **Management Tools**: Cloud management software for monitoring and resource allocation.

c) Using a diagram illustrate the IaaS Virtualization Architecture. (6 marks)

IaaS virtualization architecture is a cloud computing model that uses virtualization to provide access to computing resources. It includes networking, storage, servers, and virtualization layers.



d) Discuss the three strategies applied in virtual machine live migration techniques. (6 marks)

- **Pre-Copy Migration**:
Memory pages are copied to the destination VM while the source VM continues running. After most pages are copied, the VM is paused, and the remaining data is transferred.

- **Post-Copy Migration**:
The VM is immediately paused, and its state is moved to the destination. Memory pages are then pulled on-demand during execution.

- **Hybrid Migration**:
Combines pre-copy and post-copy methods to reduce downtime and balance performance. Memory pages are pre-copied, and any remaining pages are pulled on-demand.

e) List five key issues with virtual machine management. (5 marks)

- **Resource Allocation and Optimization**: Ensuring efficient distribution of CPU, memory, and storage among VMs.
- **Scalability**: Managing the addition of new VMs without performance degradation.
- **Security**: Protecting VMs from attacks, especially in multi-tenant environments.
- **Monitoring and Performance**: Continuously monitoring VM performance to avoid bottlenecks.
- **Migration and Downtime**: Managing live migrations without significant downtime or data loss.

f) Briefly explain 4 roles/functions of hypervisor layer. (4 marks)

- **Resource Allocation**: Allocates physical resources (CPU, memory, storage) to virtual machines.
- **Isolation**: Ensures VMs operate independently without interference.
- **Hardware Emulation**: Provides virtualized hardware for guest VMs.
- **VM Management**: Facilitates the creation, deletion, and migration of virtual machines.

QUESTION 2

a) Using an appropriate diagram, briefly explain different layers and pillars used in cloud computing reference model. (10 marks)

b) Using Mount Kenya University as the setting, compare and contrast the different cloud computing services (SAAS, PAAS, IAAS). (10 marks)

Feature	SaaS	PaaS	IaaS
Definition	Software is delivered as a service over the internet.	Provides platforms for application development and deployment.	Provides virtualized computing infrastructure.
Example at MKU	Online learning tools like Google Classroom or Zoom.	Custom application development tools for student portals.	Cloud-based storage and virtual machines for research data.
Management	Managed entirely by the provider.	Application management by MKU, infrastructure managed by the provider.	MKU manages applications and operating systems, provider manages hardware.
Customization	Limited customization of software.	Full flexibility in developing custom applications.	Custom configuration of infrastructure for specific needs.
Scalability	Scalable for user access (e.g., students).	Scales based on app development demands.	Scales based on hardware resource requirements.
Costs	Subscription-based, predictable pricing.	Pay-as-you-go for platform resources.	Pay-per-use for infrastructure.

Contrast:

- SaaS is ideal for general-purpose tools, PaaS suits custom software development, while IaaS supports infrastructure needs.
- SaaS is user-focused, PaaS targets developers, and IaaS is for system administrators.

QUESTION 3

Virtualization is the "primary ingredient" of cloud computing. Provide a detailed description of virtualization. Please include the following;

a) What is it (5 marks)

Virtualization is the process of creating a virtual version of physical resources, such as servers, storage, networks, or operating systems, by abstracting their functionalities. It allows multiple virtual environments to run on a single physical system, enabling better resource utilization, scalability, and flexibility. This technology is fundamental in cloud computing as it forms the basis for resource allocation, service delivery, and multi-tenancy.

b) What components can be virtualized (5 marks)

- **Servers**: Physical servers are divided into multiple virtual servers (virtual machines) using hypervisors.

- **Storage**: Data storage systems are abstracted into virtual storage pools accessible to multiple users.
- **Network**: Virtual networks are created using software-defined networking (SDN), enabling flexible traffic management.
- **Desktops**: Physical desktops are virtualized into virtual desktop infrastructures (VDI) accessible remotely.
- **Applications**: Applications run in isolated containers or environments to improve portability and scalability.

c) How does virtualization present itself to the end user (5 marks)

- **Virtual Machines (VMs)**: Users perceive VMs as fully functional standalone systems. For instance, cloud-hosted virtual desktops feel like personal computers.
- **Abstracted Resources**: Users access storage, applications, or platforms without knowing the physical infrastructure behind them.
- **Seamless Scaling**: End users experience automatic resource scaling, such as increasing memory or storage, based on their needs without manual adjustments.

d) Explain the resource pooling process of virtualization. (5 marks)

1. **Abstraction**: Physical hardware resources (e.g., CPU, memory, and storage) are abstracted by a virtualization layer (hypervisor).
2. **Dynamic Allocation**: Resources are assigned to virtual machines or applications based on demand, ensuring optimal utilization.
3. **Multi-Tenancy**: Multiple users or tenants share the same physical resources, with isolation to ensure security and performance.
4. **Elasticity**: Resources can be scaled up or down dynamically, depending on user requirements.
   This process underpins the scalability, cost-efficiency, and flexibility of cloud computing.

QUESTION 4

Cloud computing has become the buzzword to solve contemporary business problems. Using a sector/industry of your choice briefly describe how you would explain at least 5 benefits of cloud computing to a CIO of a company in that sector. (20 marks)

1. Cost Efficiency (4 marks)

Cloud computing eliminates the need for expensive on-premise infrastructure, such as servers and data centers. It operates on a pay-as-you-go model, ensuring the organization pays only for the resources it uses.

- **Example for healthcare**: Instead of purchasing costly equipment for patient data storage, the organization can use scalable cloud storage solutions like Amazon S3 or Google Cloud.

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2. Scalability and Flexibility (4 marks)
Cloud resources can be easily scaled up or down based on demand. This is particularly important in healthcare, where patient numbers and data requirements can fluctuate significantly.

- **Example for healthcare**: During a public health emergency, the organization can instantly scale up its storage and processing capabilities to handle increased data and patient records.

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3. Enhanced Collaboration (4 marks)
Cloud computing enables healthcare professionals to collaborate efficiently by accessing real-time data, applications, and patient records from anywhere.

- **Example for healthcare**: Doctors in different locations can access a centralized Electronic Health Record (EHR) system to provide coordinated care for a patient.

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4. Improved Data Security and Compliance (4 marks)
Cloud providers offer advanced security features, such as encryption, firewalls, and regular updates, to protect sensitive patient data. Additionally, cloud platforms comply with regulatory standards like HIPAA for healthcare data protection.

- **Example for healthcare**: By using cloud services, the CIO can ensure the security and compliance of patient information without needing an in-house security team.

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5. Disaster Recovery and Data Backup (4 marks)
Cloud computing ensures data is backed up automatically and provides robust disaster recovery solutions, minimizing downtime and data loss during system failures or cyberattacks.

- **Example for healthcare**: In the event of a ransomware attack, patient data can be quickly restored from cloud backups, ensuring continuity of care.

QUESTION 5
The Ministry of Education would like to migrate to the cloud. However, they are concerned about the security of their data once it's in the cloud. Using the principles of CIA – layout a data security plan for the Ministry of Education. (20 marks)

The **CIA Triad**—**Confidentiality**, **Integrity**, and **Availability**—provides a framework to ensure the security of data in the cloud. Below is a detailed data security plan for the Ministry of Education, addressing these principles:

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1. Confidentiality (Ensuring Data Privacy)

The Ministry must protect sensitive data, such as student records, payroll information, and examination data, from unauthorized access.
Actions:

- **Encryption**: Encrypt data at rest (e.g., AES-256 encryption) and in transit (e.g., TLS/SSL protocols) to ensure only authorized users can access it.
- **Access Controls**: Implement robust identity and access management (IAM) solutions to control who can access specific resources.
- **Multi-Factor Authentication (MFA)**: Require MFA for accessing cloud systems to add an extra layer of security.
- **Data Masking**: Mask sensitive information, such as personal student details, when used in non-production environments (e.g., training or testing).
- **Periodic Security Audits**: Regularly audit access logs to detect and respond to unauthorized access attempts.

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2. Integrity (Ensuring Data Accuracy and Consistency)

The Ministry's data must remain accurate and unaltered during storage, processing, or transmission.
Actions:

- **Checksums and Hashing**: Use hashing algorithms (e.g., SHA-256) to verify the integrity of files and ensure data has not been tampered with during transmission.
- **Version Control**: Implement versioning for important documents and databases to track changes and prevent accidental overwrites.
- **Role-Based Access**: Restrict data modification permissions to authorized personnel only, ensuring accountability.
- **Digital Signatures**: Use digital signatures to authenticate official documents and prevent forgery.
- **Regular Backups**: Conduct automated and immutable backups to restore accurate data in case of corruption or malicious activity.

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3. Availability (Ensuring Data Accessibility When Needed)

Data must be accessible to authorized users at all times, especially for critical functions like student enrollment, reporting, and resource allocation.
Actions:

- **Redundancy and Failover Systems**: Deploy redundant servers and failover mechanisms in different geographical regions to ensure minimal downtime during outages.
- **Disaster Recovery Plan**: Establish a Disaster Recovery as a Service (DRaaS) solution for quick recovery of data in case of natural disasters or cyberattacks.
- **Cloud SLAs**: Partner with cloud providers offering high availability with guaranteed uptime (e.g., 99.9%).
- **DDoS Protection**: Use Distributed Denial-of-Service (DDoS) mitigation services to prevent service disruptions caused by cyberattacks.
- **Monitoring and Alerts**: Continuously monitor cloud resources and set up alerts for potential issues, such as hardware failures or resource exhaustion.

DECEMBER 2019

QUESTION 1

a) Define cloud Computing (2 Marks)

Cloud computing refers to the delivery of computing services such as servers, storage, databases, networking, software, and more over the internet ("the cloud"), enabling users to access and store data remotely without the need for local infrastructure.

b) Outline six characteristics of cloud computing (6 Marks)

- **On-Demand Self-Service**: Users can access computing resources without requiring human intervention from the service provider.
- **Broad Network Access**: Resources are accessible over the internet through standard devices like smartphones, laptops, or desktops.
- **Resource Pooling**: Resources such as storage, processing, and memory are pooled to serve multiple users dynamically.
- **Rapid Elasticity**: Resources can be scaled up or down based on demand.
- **Measured Service**: Services are monitored, metered, and billed based on usage.
- **Multi-Tenancy**: Multiple customers share the same infrastructure while maintaining data isolation and security.

c) Outline three types of clouds (6 Marks)

- **Public Cloud**: Services are offered to the general public over the internet and are owned by a cloud service provider (e.g., AWS, Microsoft Azure).
- **Private Cloud**: Services are used exclusively by a single organization, providing greater control and security (e.g., a private cloud for a government ministry).
- **Hybrid Cloud**: Combines public and private clouds, allowing data and applications to be shared between them for flexibility and scalability.

d) Define Virtualization (2 Marks)

Virtualization is the process of creating virtual versions of physical resources, such as servers, storage devices, and networks, enabling multiple virtual machines to run on a single physical machine, optimizing resource usage.

e) Discuss three cloud service models (6 Marks)

- **Infrastructure as a Service (IaaS)**: Provides virtualized computing resources over the internet, such as virtual machines, storage, and networks. Example: Amazon EC2.
- **Platform as a Service (PaaS)**: Offers a platform that allows developers to build, test, and deploy applications without managing the underlying infrastructure. Example: Google App Engine.
- **Software as a Service (SaaS)**: Delivers software applications over the internet, accessible via web browsers. Example: Google Workspace.

f) Outline six security issues in cloud computing (6 Marks)

1. **Data Breaches**: Unauthorized access to sensitive data stored in the cloud.

1. **Data Loss**: Accidental deletion or corruption of data.
2. **Insecure Interfaces and APIs**: Vulnerabilities in APIs used to access cloud services.
3. **Denial of Service (DoS) Attacks**: Disrupting access to cloud services.
4. **Shared Technology Vulnerabilities**: Exploits arising from shared cloud infrastructure.
5. **Account Hijacking**: Unauthorized access to cloud accounts due to weak credentials or phishing.

QUESTION 2

a) Outline the components of a Cloud Service Level agreement (10 Marks)

1. **Service Availability and Uptime**: Specifies the percentage of time the service will be operational (e.g., 99.9% uptime).
2. **Performance Standards**: Defines the expected performance levels, such as response time and data processing speed.
3. **Data Security and Privacy**: Details how customer data will be protected, including encryption, access control, and compliance with regulations.
4. **Disaster Recovery and Backup**: Outlines plans for data recovery in case of disasters, including recovery time objectives (RTO) and recovery point objectives (RPO).
5. **Support and Troubleshooting**: Describes support services provided, including response times, contact methods, and issue resolution procedures.
6. **Service Scope and Responsibilities**: Clearly defines what is included in the service and the responsibilities of both the provider and the customer.
7. **Change Management**: Establishes rules for updating or modifying the service, including notification periods.
8. **Penalties and Remedies**: Specifies penalties for non-compliance with SLA terms, such as credits or refunds for downtime.
9. **Monitoring and Reporting**: Details how the service provider will monitor performance and report on SLA compliance.
10. **Exit Strategy and Data Portability**: Provides procedures for terminating the agreement, including data retrieval and transition to another provider.

b) Discuss the benefits of Infrastructure-as-a-Service (10 Marks)

- **Scalability**: IaaS allows businesses to scale resources up or down based on demand, ensuring flexibility without over-provisioning.
- **Cost-Effectiveness**: Reduces capital expenditure by providing virtual infrastructure on a pay-as-you-go basis, minimizing upfront investments.
- **Quick Deployment**: Enables rapid setup of IT environments without the need for physical hardware, speeding up deployment.
- **Global Reach**: Allows access to resources from anywhere, supporting global operations and remote work.
- **Disaster Recovery and Backup**: Built-in disaster recovery solutions ensure data and service continuity during failures.
- **Focus on Core Business**: By offloading infrastructure management to the provider, businesses can focus on their core competencies.
- **Improved Security**: IaaS providers implement advanced security measures like encryption and intrusion detection.
- **Resource Optimization**: Maximizes utilization of resources by virtualizing hardware, reducing waste.
- **Innovation and Experimentation**: Provides a platform for testing new technologies or applications without risking large investments.
- **Access to Advanced Technology**: Offers the latest hardware and software technologies without the need for continuous updates or upgrades.

QUESTION 3

Discuss the application of cloud computing solution in the following areas;

(i) Energy Systems (5 Marks)

- **Smart Grids**: Cloud computing enables the management of smart grids by processing vast amounts of real-time data for energy distribution and consumption optimization.
- **Energy Forecasting**: Facilitates accurate prediction of energy demands and renewable energy production through data analytics.
- **Remote Monitoring**: Allows utilities to monitor and manage infrastructure such as power plants and wind turbines remotely.
- **Cost Management**: Provides tools for analyzing energy usage patterns and optimizing operational costs.
- **Sustainability**: Cloud platforms help integrate renewable energy sources, reducing reliance on fossil fuels.

(ii) Transportation Systems (5 Marks)

1. **Traffic Management**: Cloud solutions support real-time traffic analysis and optimization, reducing congestion.
2. **Fleet Management**: Enables tracking, monitoring, and optimization of transportation fleets using GPS and IoT integration.
3. **Passenger Services**: Provides cloud-based ticketing and scheduling systems for airlines, trains, and buses.
4. **Autonomous Vehicles**: Processes large datasets for AI models in self-driving cars, enabling real-time decision-making.

5. **Predictive Maintenance**: Uses cloud analytics to predict and prevent failures in transportation equipment, improving safety and reducing downtime.

(iii) Manufacturing Industry (5 Marks)

1. **Smart Factories**: Facilitates IoT and automation in manufacturing processes for improved efficiency and productivity.
2. **Supply Chain Management**: Enhances supply chain visibility and coordination using real-time data analytics.
3. **Quality Control**: Cloud-based systems analyze production data to identify and address quality issues.
4. **Product Design and Development**: Provides collaborative tools and platforms for rapid prototyping and testing of new designs.
5. **Energy Optimization**: Monitors and reduces energy consumption in factories through cloud-based energy management systems.

(iv) Healthcare (5 Marks)

- **Electronic Health Records (EHRs)**: Cloud systems store and manage patient records securely, allowing seamless access by healthcare professionals.
- **Telemedicine**: Enables remote consultations and diagnoses using cloud-hosted platforms.
- **Medical Research**: Facilitates collaboration and analysis of large datasets for drug discovery and clinical trials.
- **Healthcare Analytics**: Uses data analytics to predict patient outcomes, manage diseases, and optimize hospital operations.
- **Disaster Recovery**: Ensures critical patient data and healthcare services remain available during disasters or emergencies.

QUESTION 4

You have been appointed to the Meru University of Science and Technology as the Online Education Systems Administrator. Discuss the solutions that you are likely to deploy to enable Cloud Computing for Education. (20 Marks)

1. Learning Management Systems (LMS) in the Cloud (4 Marks)

Deploy a cloud-based LMS (e.g., Moodle Cloud, Google Classroom) to provide a centralized platform for course delivery, student engagement, and content sharing.

The solution would support online classes, assignments, quizzes, and grading, ensuring 24/7 accessibility for students and lecturers.

2. Cloud Storage Solutions (4 Marks)

Implement secure cloud storage solutions (e.g., Google Drive, Microsoft OneDrive, or AWS S3) to store academic materials, research documents, and student records.

These solutions allow students and staff to collaborate and access resources from anywhere, reducing the risk of data loss.

3. Virtual Classrooms and Collaboration Tools (3 Marks)
Integrate platforms like Zoom, Microsoft Teams, or Google Meet for hosting virtual lectures, discussions, and webinars.
These tools enable real-time interaction, breakout rooms for group work, and live streaming for lectures.
4. Cloud-Based Research Platforms (3 Marks)
Deploy research environments in the cloud using platforms like IBM Cloud or Google Cloud for Education to provide computational power and tools for research and innovation.
This would support data analysis, machine learning, and simulations for students and faculty conducting advanced research.
5. Disaster Recovery and Backup Solutions (2 Marks)
Utilize cloud disaster recovery solutions to ensure academic records, research data, and administrative systems are protected and recoverable in case of cyber-attacks or system failures.
6. Student Information Systems (SIS) in the Cloud (2 Marks)
Migrate the university's SIS to the cloud for managing student enrollment, academic progress, and financial records. This ensures scalability and real-time access to data.
7. Cloud-Based Email and Productivity Tools (2 Marks)
Provide email and productivity tools (e.g., G Suite for Education or Microsoft 365) to students and staff, supporting communication, file sharing, and collaborative projects.
8. Content Delivery Networks (CDN) (2 Marks)
Implement CDN services to improve the speed and reliability of delivering online content (e.g., lecture videos and academic materials) to students across the region or globe.

QUESTION 5
Cloud Security Alliance (CSA) provides a Trusted Clound Initiative (TCI) Reference Architecture. Discuss the Security Risk Management (SRM) domain within the TCI Reference. (20 Marks)

The **Security Risk Management (SRM)** domain within the Cloud Security Alliance (CSA) Trusted Cloud Initiative (TCI) Reference Architecture focuses on identifying, analyzing, mitigating, and monitoring security risks in cloud environments. This domain ensures that cloud service providers (CSPs) and customers adopt practices that minimize vulnerabilities, protect assets, and maintain compliance with security standards. Below is a detailed discussion of the SRM domain:

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1. Risk Identification and Assessment

- The SRM domain starts with identifying and assessing risks related to cloud adoption.
- It includes evaluating risks specific to the **cloud delivery models** (IaaS, PaaS, SaaS) and **deployment types** (public, private, hybrid, and community clouds).

- Organizations must analyze potential threats, vulnerabilities, and impacts on confidentiality, integrity, and availability (CIA) of data.

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2. Cloud Risk Framework

- The SRM domain uses industry-standard frameworks such as **NIST Risk Management Framework (RMF)** or **ISO 31000** for consistent and structured risk evaluation.
- These frameworks help classify risks based on their likelihood and potential impact.

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3. Threat Modeling in Cloud Environments

- Threat modeling involves identifying threats specific to multi-tenant cloud systems, shared infrastructure, APIs, and data transit.
- Common threats include **data breaches, insider threats, insecure interfaces, and advanced persistent threats (APTs)**.

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4. Risk Mitigation Strategies

- The domain defines mitigation measures to reduce risks, such as:
    - **Data Encryption**: Ensuring data is encrypted at rest, in transit, and in use.
    - **Access Control**: Implementing least privilege and multi-factor authentication (MFA).
    - **Network Security**: Using firewalls, intrusion detection/prevention systems (IDS/IPS), and secure VPNs.
    - **Continuous Monitoring**: Deploying tools to monitor anomalies and potential threats in real time.

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5. Compliance and Legal Risk Management

- Ensures compliance with regulatory and legal frameworks such as GDPR, HIPAA, or PCI DSS.
- The domain addresses risks associated with data residency, jurisdiction, and contracts with CSPs.

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6. Incident Management and Response

- Establishes a structured approach to **detecting, responding to, and recovering from security incidents** in the cloud.

- Includes creating **incident response playbooks**, ensuring the availability of forensic tools, and defining roles for stakeholders.

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7. Risk Communication and Reporting

- The SRM domain emphasizes effective communication of identified risks and mitigations to stakeholders, including cloud providers, customers, and regulators.
- Risk reports are shared to ensure transparency and informed decision-making.

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8. Continuous Risk Monitoring and Auditing

- Implements tools and processes to continuously monitor risks in dynamic cloud environments.
- Includes regular auditing of security controls and conducting **penetration testing** to ensure vulnerabilities are addressed proactively.

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9. Key Areas Addressed by SRM

- **Data Security**: Risks related to data breaches, loss, or unauthorized access.
- **Identity and Access Management (IAM)**: Risks of unauthorized access due to weak credentials or excessive permissions.
- **Cloud Supply Chain Risks**: Ensuring third-party providers and vendors comply with security standards.
- **Shared Responsibility Model**: Addressing ambiguities in responsibilities between CSPs and customers for security controls.

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10. Practical Example

For instance, in a hybrid cloud scenario where a university stores sensitive research data in the cloud, the SRM domain ensures:

- Risk assessment of third-party cloud providers.
- Encryption of research data to prevent breaches.
- IAM policies to limit access only to authorized researchers.
- Compliance with local and international data protection laws.

AUGUST 2021
QUESTION 1

a) Cloud computing relies on virtualization technology; giving examples in each, differentiate the two basic types of virtualization (6 marks)

- **Type 1 Virtualization (Bare-Metal Virtualization)**

  - Installed directly on the physical hardware.
  - Examples: VMware ESXi, Microsoft Hyper-V, XenServer.
  - It provides high performance and is typically used in enterprise environments.

- **Type 2 Virtualization (Hosted Virtualization)**

  - Installed on top of an operating system.
  - Examples: Oracle VirtualBox, VMware Workstation, Parallels Desktop.
  - Easier to set up but less efficient compared to Type 1.

b) Outline five characteristics of cloud computing (5 marks)

1. **On-Demand Self-Service**: Users can provision computing resources as needed without human interaction with the provider.
2. **Broad Network Access**: Resources are accessible over the internet using various devices.
3. **Resource Pooling**: Multiple users share pooled computing resources dynamically allocated based on demand.
4. **Rapid Elasticity**: Resources can be scaled up or down automatically to meet user needs.
5. **Measured Service**: Resource usage is monitored and billed based on a pay-as-you-go model.

c) Distinguish the following terminologies with relation to Cloud Computing;

i) Private cloud and Public Cloud (4 marks)

- **Private Cloud**: Exclusive to one organization, providing higher control and security. Example: A bank using its own cloud for financial transactions.
- **Public Cloud**: Open to the public, offered by third-party providers. Example: Services like Google Cloud and AWS.

ii) BPaaS and INaaS (4 marks)

- **BPaaS (Business Process as a Service)**: Focuses on outsourcing business processes like payroll or accounting to the cloud.
- **INaaS (Integration as a Service)**: Provides tools to integrate various systems and data sources across cloud platforms.

d) To offset the price you pay for the cloud service, you need to realize a commensurate benefit; discuss three utility price models (6 marks)

- **Pay-As-You-Go**: Users pay based on actual resource usage (e.g., per hour of compute time or per GB of storage).

- **Subscription-Based**: Fixed monthly or annual fees for pre-determined resource levels, regardless of usage.
- **Spot Pricing**: Dynamic pricing where users bid for spare capacity at reduced costs. Commonly used in non-critical workloads.

e) List five disadvantages of cloud computing (5 marks)

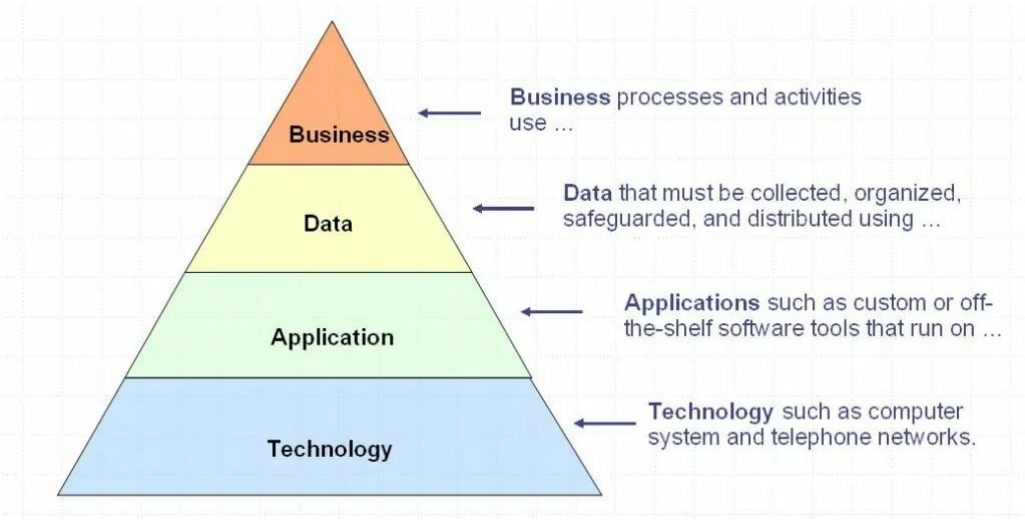
- **Downtime**: Reliance on the internet means services can become unavailable during outages.
- **Security Risks**: Data in the cloud is vulnerable to breaches and unauthorized access.
- **Limited Control**: Users have limited control over the underlying infrastructure and data management.
- **Hidden Costs**: Long-term costs for data storage and retrieval may exceed initial expectations.
- **Compliance Challenges**: Ensuring compliance with laws and regulations across different regions can be complex.

QUESTION 2

a) IT Enterprise architecture recognizes four architecture domains. Using a diagram outline the layers and their functions. (8 marks)

The **four architecture domains** in IT enterprise architecture are:

1. **Business Architecture**: Defines business processes, organizational structures, and strategies.
2. **Data Architecture**: Focuses on data management, including storage, integration, and security.
3. **Application Architecture**: Describes software applications and how they interact with business processes and data.
4. **Technology Architecture**: Provides the infrastructure (hardware, networks, and platforms) required to support applications and data.



b) Describe the five key properties of Cloud computing (6 marks)

- **On-Demand Self-Service**: Users can provision and manage resources as needed without requiring provider interaction.
- **Broad Network Access**: Resources are accessible over the internet via various devices (e.g., smartphones, laptops).
- **Resource Pooling**: Resources are shared among multiple users, dynamically allocated and reallocated based on demand.
- **Rapid Elasticity**: Resources can scale up or down automatically to meet workload demands.
- **Measured Service**: Cloud usage is monitored, controlled, and billed based on metrics like storage or processing time.

c) A service broker acts as an intermediary between a service consumer and a number of service providers; briefly discuss the three roles that the broker plays. (6 marks)

1. **Service Discovery**: Helps users identify and select suitable cloud services based on their needs.
2. **Service Integration**: Facilitates the integration of services from multiple providers to work seamlessly together.
3. **Service Management**: Manages service provisioning, monitoring, and billing on behalf of the consumer.

<u>QUESTION 3</u>

a) On-demand self-service has the advantage of empowering you to consume the service whenever you want; highlight the two characteristics of availability. (6 marks)

- **High Availability**: The service remains operational and accessible for the majority of the time, often measured by uptime percentages (e.g., 99.99%).
- **Fault Tolerance**: The system can continue functioning even when there are failures in hardware, software, or network components, ensuring uninterrupted access.

b) Briefly explain the following with relation to cloud computing;

i) SaaS (2 marks)

Software as a Service delivers software applications over the internet. Users access these applications via a web browser without managing infrastructure or installations (e.g., Google Workspace).

ii) PaaS (2 marks)

Platform as a Service provides a platform and environment to develop, test, and deploy applications. It includes tools and libraries to support development (e.g., Microsoft Azure App Service).

iii) IaaS (2 marks)

Infrastructure as a Service offers virtualized computing resources such as servers, storage, and networking over the internet. It allows users to manage operating systems and applications (e.g., Amazon EC2).

iv) DBaas (3 marks)

Database as a Service provides managed database services where users can store, retrieve, and manipulate data without handling database management tasks like updates, scaling, or backups (e.g., AWS RDS).

c) In 2014 there were about three million data centers in the U.S. While other computing

paradigms have failed, cloud computing has continued to be successful. List five reasons for

the success of cloud computing (5 marks)

Cost Efficiency: Reduces the need for capital expenditure on infrastructure.

Scalability: Allows businesses to easily scale resources based on demand.

Flexibility: Enables access to services from any location via the internet.

Innovation: Encourages rapid development and deployment of applications.

Reliability: Offers high availability, fault tolerance, and disaster recovery options.

QUESTION 4

a) Cloud delivery model has gained popularity in the recent past with regards to deployment.

Discuss the new philosophical attributes for delivering computing services (6 marks)

Resource Pooling: Centralized resources are shared among multiple users using multi-tenancy, allowing better utilization and flexibility.

On-Demand Self-Service: Users can provision resources like servers, storage, or applications as needed without requiring human intervention.

Pay-As-You-Go: Users only pay for the resources they use, eliminating the need for upfront capital investment.

Elasticity and Scalability: Resources can scale up or down automatically to meet workload demands.

Broad Network Access: Services are accessible over the internet, enabling remote and multi-device access.

Measured Service: Usage is monitored, controlled, and reported to provide transparency and optimize resource utilization.

b) Cloud computing has resulted to a paradigm shift from local to network-centric data processing and storage; discuss the services of concern and benefits (8 marks)

Services of Concern:

1. **Storage as a Service**: Provides reliable and scalable cloud storage for data backups, archives, and sharing.
2. **Compute as a Service**: Offers virtualized computing power for running applications and workloads without owning physical infrastructure.
3. **Database as a Service (DBaaS)**: Managed database services to handle structured and unstructured data efficiently.

Benefits:

1. **Cost Reduction**: Eliminates capital expenditure on hardware and reduces operational costs.
2. **Scalability**: Allows businesses to scale resources up or down based on demand.
3. **Accessibility**: Enables remote access to applications and data from anywhere, improving collaboration.
4. **Business Continuity**: Provides reliable disaster recovery and data backup solutions to minimize downtime.
5. **Innovation**: Facilitates quick deployment and development of new applications.

c) Describe three popular cloud computing resources (6 marks)

- **Virtual Machines (VMs)**: Virtualized computing environments that simulate physical hardware, enabling users to run applications and operating systems independently.
- **Storage**: Cloud-based storage solutions (e.g., Amazon S3, Google Cloud Storage) that allow users to store, access, and manage data on demand.
- **Databases**: Managed database services (e.g., AWS RDS, Azure SQL) for structured and unstructured data storage, providing scalability, security, and ease of use.

QUESTION 5

a) Cloud computing is based on a paradigm shift with profound implications on computing ethics. Outline the three main elements of this shift (6 marks)

- **Decentralized Ownership and Control**: Data and computing resources are no longer managed solely by the organization but are hosted and controlled by third-party cloud service providers, raising concerns about data privacy and sovereignty.
- **Shared Resources**: The multi-tenant model of cloud computing involves sharing infrastructure among multiple users, which can lead to ethical concerns about data segregation, unauthorized access, and accountability.

- **Global Access and Connectivity**: Cloud services provide worldwide access, increasing the risk of ethical dilemmas related to jurisdiction, censorship, and misuse of services in different legal and cultural contexts.

b) An attack at Akamai on June 15, 2004 caused a domain name outage and a major blackout that affect Google, Yahoo and many other sites. Giving actual real life examples, briefly discuss four cloud vulnerabilities and how the affected organizations recovered from the vulnerabilities. (8 marks)

1. **Denial of Service (DoS) Attacks**
    - *Example*: The Akamai outage caused by a massive DoS attack disrupted services for major companies.
    - *Recovery*: Companies worked with Akamai to strengthen their DNS infrastructure and implemented advanced traffic filtering systems to mitigate future attacks.
2. **Data Breaches**
    - *Example*: In 2019, Capital One experienced a breach exposing over 100 million customer records due to a cloud misconfiguration.
    - *Recovery*: The organization revamped its cloud security policies, conducted regular audits, and adopted advanced monitoring tools.
3. **Insider Threats**
    - *Example*: Tesla suffered an insider threat in 2018 when an employee intentionally sabotaged cloud configurations, leading to disruptions.
    - *Recovery*: Tesla implemented stricter access controls, monitored user activities, and enhanced internal auditing processes.
4. **Shared Technology Vulnerabilities**
    - *Example*: The Spectre and Meltdown vulnerabilities in 2018 affected shared virtualized environments in cloud services.
    - *Recovery*: Organizations applied vendor-released patches, updated hardware and software configurations, and optimized security in shared environments.

c) As a cloud user, you are faced with challenges ranging from security, compliance, managing cost and lack of expertise. Discuss the measures you will attempt to reduce the cloud costs.

(6 marks)

- **Right-Sizing Resources**: Analyze workloads and scale resources appropriately to avoid paying for unused capacity.
- **Leverage Reserved Instances**: Opt for long-term contracts with providers (e.g., AWS Reserved Instances) for predictable workloads to benefit from discounted rates.
- **Monitor and Optimize Usage**: Use monitoring tools to track cloud usage and eliminate redundant resources, such as idle virtual machines or unused storage.
- **Use Auto-Scaling**: Configure auto-scaling to match resource allocation with demand dynamically, reducing costs during low usage periods.

- **Take Advantage of Free Tiers**: Use the free-tier services offered by cloud providers for testing and development purposes.
- **Consider Multi-Cloud Strategies**: Compare pricing across providers and use services from different vendors to optimize cost efficiency.

OCTOBER 2022

QUESTION 1

a) Using appropriate examples, describe three security features that devices that access the

cloud should have (6 marks)

Encryption: Devices must use encryption to secure data in transit and at rest. For example, devices accessing cloud storage like Google Drive should utilize HTTPS or SSL/TLS protocols to ensure secure data transmission.

Authentication Mechanisms: Multi-Factor Authentication (MFA) ensures that only authorized users access cloud services. For instance, MFA for accessing Microsoft OneDrive combines a password and a one-time code sent to a mobile device.

Endpoint Security Software: Devices must have antivirus and anti-malware software to protect against malicious attacks that could compromise cloud data. For example, using endpoint protection software like Symantec Endpoint Security.

b) Provide four major reasons why cloud services might become unavailable (4 marks)

Network Connectivity Issues: Loss of internet connection can disrupt access to cloud services.

Data Center Outages: Physical infrastructure failures, such as power outages or server crashes, can lead to service downtime.

Denial of Service (DoS) Attacks: Cyberattacks that flood cloud servers with traffic can render services inaccessible.

Cloud Provider Maintenance: Scheduled updates or unanticipated maintenance by the provider can temporarily affect service availability.

c) Briefly explain three roles/functions of hypervisor layer (6 marks)

Resource Allocation: The hypervisor dynamically allocates resources (CPU, memory, and storage) to virtual machines (VMs) as needed.

Isolation: It ensures VMs are isolated from one another, preventing interference or security breaches across VMs.

Hardware Emulation: The hypervisor emulates hardware for VMs, allowing them to run independently of the underlying physical hardware.

d) You have been identified as an expert in cloud computing. You are expected to share a

talk on software as a service (SaaS). In your answer, describe two SaaS maturity models.

Illustrate your answer with suitable examples (4 marks)

Ad Hoc/Custom: SaaS is built to meet specific customer requirements with limited scalability. Example: A bespoke project management software tailored for a single enterprise.

Configurable: SaaS solutions are highly configurable to meet varying customer needs while still being standardized. Example: Salesforce CRM, which allows users to customize workflows, fields, and reports.

e) Highlight are the two I/O virtualization subsystem architecture (2 marks)

Pass-Through I/O: The virtual machine directly accesses the physical hardware device.

Shared I/O: The hypervisor mediates and shares physical I/O devices among multiple virtual machines.

f) List three emulation implementation techniques (3 marks)

Full Emulation: Mimics hardware entirely in software.

Paravirtualization: Modifies the guest OS to interact directly with the hypervisor.

Binary Translation: Converts guest OS instructions into instructions compatible with the host system.

g) Differentiate the two types of virtual machine hypervisors (2 marks)

Type 1 (Bare-Metal Hypervisor): Runs directly on physical hardware without an operating system. Example: VMware ESXi.

Type 2 (Hosted Hypervisor): Runs on top of an existing operating system. Example: Oracle VirtualBox.

h) Discuss the three hardware virtualization techniques (3 marks)

Full Virtualization: The VM simulates a complete hardware environment, allowing unmodified guest OS to run.

Paravirtualization: The guest OS is modified to work alongside the hypervisor, improving performance by reducing overhead.

Hardware-Assisted Virtualization: Relies on CPU features like Intel VT or AMD-V to support efficient virtualization without OS modification.

QUESTION 2

a) Define cloud governance? Using appropriate examples, briefly describe five reasons why cloud governance is important (10 marks)

Cloud governance refers to the set of rules, policies, and procedures that ensure proper management, security, and efficient use of cloud resources. It provides a framework for organizations to align their cloud usage with business objectives while maintaining control over costs, security, and compliance.

Reasons why cloud governance is important:

1. **Cost Optimization**: Cloud governance helps track resource usage and manage spending, preventing wastage by ensuring only necessary resources are allocated. For example, applying policies to automatically shut down idle virtual machines.
2. **Data Security and Compliance**: It ensures compliance with industry standards and regulations like GDPR or HIPAA by defining how data is stored and accessed securely.
3. **Risk Management**: Governance identifies potential risks such as unauthorized access and provides measures to mitigate them, such as access control policies.
4. **Operational Efficiency**: Streamlines cloud operations by standardizing configurations, resource provisioning, and management processes.
5. **Accountability and Transparency**: Provides clear roles, responsibilities, and audit trails for cloud usage, ensuring accountability and visibility. For instance, logging every API call in AWS CloudTrail for security audits.

b) Cloud Component Model (CCM) approach for application design involves three steps elaborate (6 marks)

- **Component Identification**:
Identify and analyze the various components of an application, such as compute, storage, and networking, required for deployment in the cloud.

- **Component Modeling**:
Design reusable cloud-native components by defining their relationships, dependencies, and interactions. This involves abstracting components for modular design.

- **Component Deployment**:
Deploy the designed components into the cloud environment while ensuring scalability, resilience, and performance. This includes selecting the appropriate cloud service model (IaaS, PaaS, SaaS).

c) Outline the Model View Controller (MVC) design pattern for web applications (4 marks)

- **Model**:

Represents the data and business logic of the application. It manages the database interactions and handles data processing. For example, a database schema or API logic.

- **View**:

The presentation layer that displays data to users and captures user input. It is responsible for rendering UI elements like HTML pages.

- **Controller**:

Acts as an intermediary between the Model and the View. It processes user requests, retrieves data from the Model, and passes it to the View. For example, a controller handling form submissions and displaying the results.

QUESTION 3

a) Outline the three cloud computing layers (6 marks)

1. **Software as a Service (SaaS):**
   Provides applications and software to end-users over the internet. Users access these services via web browsers without managing underlying infrastructure. Examples: Gmail, Microsoft 365, Salesforce.
2. **Platform as a Service (PaaS):**
   Offers a platform for developers to build, test, and deploy applications without managing hardware or software infrastructure. Examples: Google App Engine, AWS Elastic Beanstalk.
3. **Infrastructure as a Service (IaaS):**
   Provides virtualized computing resources such as servers, storage, and networking on demand. Users manage applications and operating systems while the provider handles hardware. Examples: AWS EC2, Microsoft Azure, Google Cloud Compute.

b) Describe the application and infrastructure focused cloud computing layers services (6 marks)

- **Application-Focused Layer (SaaS):**

  - Offers end-user applications delivered over the internet.
  - Examples of services include customer relationship management (CRM), enterprise resource planning (ERP), and collaboration tools.
  - Service providers handle application updates, security, and data storage.

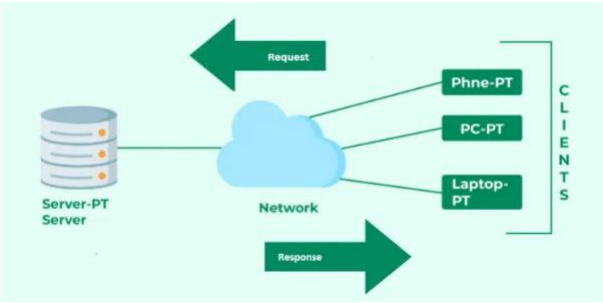
- **Infrastructure-Focused Layer (IaaS):**

  - Delivers virtualized resources such as servers, storage, and networking.
  - Allows businesses to deploy and run operating systems, applications, and middleware.
  - Provides scalability and flexibility to build custom IT environments.

c) Using a diagram differentiate client/server and cloud architectures (8 marks)

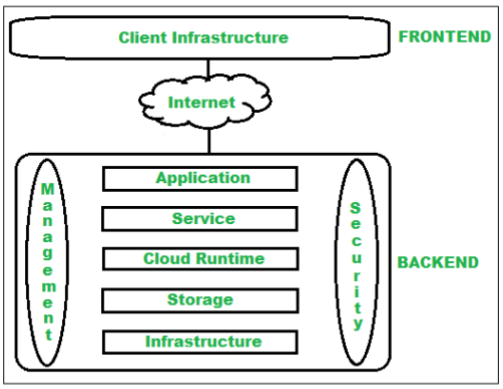
- **Client/Server Architecture:**

  - Centralized server hosting applications, with clients (users) connecting via a network.
  - Users are dependent on the server's capacity and availability.
  - Examples: Traditional LAN-based applications like file servers or databases.



- **Cloud Architecture:**

  - Resources and applications are hosted in distributed cloud data centers.
  - Access is available over the internet, offering scalability and reliability.
  - Examples: Google Drive, AWS, Dropbox.



QUESTION 4

a) Using a diagram illustrate the two modes of CPU architecture (6 marks)

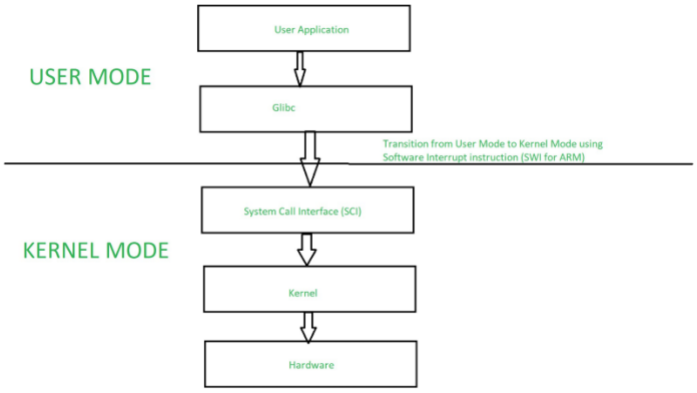
User Mode:

- Limited access to hardware resources.
- Used for executing user applications.
- Transition to kernel mode occurs via system calls.

In a CPU architecture, "kernel mode" refers to a privileged execution state where the operating system kernel runs with unrestricted access to all system resources and hardware.

Kernel Mode:

- Full access to hardware resources.
- Used for executing OS-level processes like hardware management.



b) Highlight three goals of memory virtualization (6 marks)

Isolation: Ensures that processes are separated from each other to prevent interference or access to another process's memory.

Efficiency: Optimizes memory usage by allowing multiple virtual machines (VMs) to share the same physical memory.

Flexibility: Enables dynamic allocation and reallocation of memory to meet changing workload requirements.

c) Describe the limiting factors for virtual machine on multi-core CPUs (8 marks)
Resource Contention: Multiple VMs sharing the same CPU cores can lead to contention, reducing overall performance.

Cache Coherency Issues: Sharing cache between VMs can cause latency due to frequent updates and synchronization overhead.

NUMA (Non-Uniform Memory Access) Challenges: Poor VM placement across NUMA nodes can lead to higher memory access latency.

I/O Bottlenecks: High I/O demand from VMs can saturate the host's input/output capacity.

Scheduler Overhead: The hypervisor's CPU scheduling for VMs can introduce latency and inefficiencies in multi-core environments.

Scaling Limitations: Increasing the number of cores assigned to a single VM may not result in linear performance improvement due to inter-core communication overhead.

QUESTION 5

Discuss the application of cloud computing solution in the following areas;

(i) Energy Systems (5 Marks)

- **Smart Grids**: Cloud computing enables the management of smart grids by processing vast amounts of real-time data for energy distribution and consumption optimization.
- **Energy Forecasting**: Facilitates accurate prediction of energy demands and renewable energy production through data analytics.
- **Remote Monitoring**: Allows utilities to monitor and manage infrastructure such as power plants and wind turbines remotely.
- **Cost Management**: Provides tools for analyzing energy usage patterns and optimizing operational costs.
- **Sustainability**: Cloud platforms help integrate renewable energy sources, reducing reliance on fossil fuels.

(ii) Transportation Systems (5 Marks)

6. **Traffic Management**: Cloud solutions support real-time traffic analysis and optimization, reducing congestion.
7. **Fleet Management**: Enables tracking, monitoring, and optimization of transportation fleets using GPS and IoT integration.
8. **Passenger Services**: Provides cloud-based ticketing and scheduling systems for airlines, trains, and buses.
9. **Autonomous Vehicles**: Processes large datasets for AI models in self-driving cars, enabling real-time decision-making.
10. **Predictive Maintenance**: Uses cloud analytics to predict and prevent failures in transportation equipment, improving safety and reducing downtime.

(iii) Manufacturing Industry (5 Marks)

6. **Smart Factories**: Facilitates IoT and automation in manufacturing processes for improved efficiency and productivity.
7. **Supply Chain Management**: Enhances supply chain visibility and coordination using real-time data analytics.
8. **Quality Control**: Cloud-based systems analyze production data to identify and address quality issues.
9. **Product Design and Development**: Provides collaborative tools and platforms for rapid prototyping and testing of new designs.
10. **Energy Optimization**: Monitors and reduces energy consumption in factories through cloud-based energy management systems.

(iv) Healthcare (5 Marks)

- **Electronic Health Records (EHRs)**: Cloud systems store and manage patient records securely, allowing seamless access by healthcare professionals.
- **Telemedicine**: Enables remote consultations and diagnoses using cloud-hosted platforms.
- **Medical Research**: Facilitates collaboration and analysis of large datasets for drug discovery and clinical trials.
- **Healthcare Analytics**: Uses data analytics to predict patient outcomes, manage diseases, and optimize hospital operations.
- **Disaster Recovery**: Ensures critical patient data and healthcare services remain available during disasters or emergencies.

DECEMBER 2023

QUESTION 1

(a) Cloud computing can offer a number of service models. Explain Two benefits and Three application areas of Identity-as-a-service model(Iaas) [5 marks]

Two benefits:

1. **Centralized Identity Management:** IDaaS enables efficient and secure management of user identities, authentication, and access control across multiple applications and systems.
2. **Improved Security:** Reduces risks by enforcing strong authentication mechanisms like multi-factor authentication (MFA) and enabling secure Single Sign-On (SSO).

Three application areas:

1. **Access Management:** Managing user access to cloud and on-premise applications.
2. **User Authentication:** Providing MFA and SSO for secure login processes.
3. **Compliance Monitoring:** Ensuring adherence to regulatory requirements like GDPR by auditing user access and activity.

(b) Explain five Characteristics of Cloud Computing [5marks]

- **On-Demand Self-Service:** Users can access computing resources as needed without requiring manual intervention.
- **Broad Network Access:** Resources are accessible over the internet using standard devices such as laptops, phones, and tablets.
- **Resource Pooling:** Resources like storage and computing power are shared among multiple users dynamically.
- **Scalability and Elasticity:** Cloud services can scale up or down based on user demand.
- **Measured Service:** Cloud usage is metered, and users are billed based on their consumption (pay-as-you-go model).

(c) Highlight the use of FOUR components maintained in a node of Google cloud platform. [4 marks]

- **Compute Engine:** Handles virtual machine instances for running applications.
- **Cloud Storage:** Manages data storage needs, providing scalable and durable storage solutions.
- **BigQuery:** Provides data analytics and querying capabilities for large datasets.
- **Kubernetes Engine:** Manages containerized applications and workloads.

(d) Highlight three goals of memory virtualization [6marks]

1. **Isolation:** Separates memory spaces of different virtual machines to ensure security and independence.
2. **Efficiency:** Maximizes resource utilization by allowing multiple VMs to share physical memory effectively.
3. **Flexibility:** Enables dynamic allocation and reallocation of memory based on workload demands.

(e) Distinguish between the following computations paradigms.

a. Centralized vs. distributed computing [4 marks]

- **Centralized Computing:** All processing and storage are performed on a single central server or system. Example: Mainframe systems.
- **Distributed Computing:** Processing is spread across multiple interconnected systems, enhancing fault tolerance and scalability. Example: Blockchain networks.

b. Parallel vs. cloud computing [4 marks]

- **Parallel Computing:** Tasks are divided into smaller parts and executed simultaneously on multiple processors within the same machine. Example: Supercomputers.
- **Cloud Computing:** Provides on-demand network access to shared resources (e.g., servers, storage) over the internet. Example: AWS, Google Cloud.

(f) Name FOUR providers of cloud computing services [2 marks]

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform
- IBM Cloud

QUESTION 2

(a) Explain how the following technologies form part of the enablers of cloud computing: -

i) Web 2.0 [2 marks]

Web 2.0 technologies enable dynamic, interactive, and user-friendly web applications, allowing cloud users to easily interact with services through interfaces like browsers, making services such as SaaS accessible and functional.

ii) Distributed Storage [2 marks]

Distributed storage systems provide scalable, reliable, and redundant storage for cloud data across multiple servers and locations, ensuring data availability and fault tolerance.

iii) Utility Computing [2 marks]

Utility computing provides a pay-as-you-go model for accessing computing resources (e.g., processing power and storage) as needed, forming the economic basis of cloud computing.

iv) Network Bandwidth & Latency [2 marks]

High bandwidth and low latency are critical for seamless access to cloud services, enabling real-time data transfer and interaction with cloud-based applications.

v) Fault-Tolerant Systems [2 marks]

Fault-tolerant systems ensure continuous operation and high availability of cloud services by replicating resources and automatically handling failures in hardware or software.

(b) Explain five security challenges of cloud computing [5 marks]

Data Breaches: Unauthorized access to sensitive data stored in the cloud.

Insecure APIs: Poorly secured interfaces for interacting with cloud services can be exploited by attackers.

Data Loss: Accidental or malicious deletion of data without proper backups.

Shared Responsibility Risks: Misunderstanding the division of security roles between cloud providers and users can lead to vulnerabilities.

Insider Threats: Malicious or negligent actions by employees with access to cloud systems can compromise security.

(c) Using a block diagram, differentiate between front end and backend of cloud computing,

and the services they can offer. [5 marks]

QUESTION 3

(a) With reference to hardoop core in cloud computing, answer the following questions: -

i) What is hardoop core in cloud computing [2 marks]

Hadoop Core refers to the fundamental components of the Apache Hadoop framework, which includes HDFS (Hadoop Distributed File System) for distributed storage and MapReduce for distributed data processing. It allows efficient handling of large-scale data across multiple nodes in a cloud environment.

ii) Explain Four steps of Map-reduce in hardoop core [4 marks]

MapReduce follows these key steps:

1. **Splitting (Input Splitting)** – The input data is divided into smaller chunks for parallel processing.
2. **Mapping** – Each split is processed by a map function to generate intermediate key-value pairs.
3. **Shuffling & Sorting** – The intermediate key-value pairs are grouped and sorted based on the key.
4. **Reducing** – The grouped data is processed using the reduce function to generate the final output.

iii) Describe the need for HDFS (hardoop distributed file system) in cloud computing

[4 marks]

Fault Tolerance – HDFS replicates data across multiple nodes, ensuring data availability even if a node fails.

Scalability – It supports distributed storage across multiple machines, allowing large-scale data management.

Cost-Effectiveness – Uses commodity hardware, reducing infrastructure costs.

High Throughput – Optimized for batch processing, ensuring efficient data retrieval and storage.

(b) Computer virtualization has really assisted various computing technologies such as cloud

computing, discuss. [4 marks]

Computer virtualization enables cloud computing by:

Resource Optimization – Virtual machines (VMs) allow multiple OS instances to run on a single physical server.

Scalability – Virtualized environments can scale resources dynamically based on demand.

Cost Efficiency – Reduces hardware costs by maximizing server utilization.

Isolation & Security – Each VM operates independently, enhancing security and system stability.

(c) Using sketch diagrams, explain the following distributed computing paradigms

i) Edge computing [3 marks]

ii) Grid computing [3 marks]

QUESTION 4

a) With the help of a diagram, explain storage grid virtualization in cloud computing. [6 marks]

b) With reference to Network-as-a-Service, answer the following questions:-

i) What is Network-as-a-service (NaaS) [2 marks]

NaaS is a cloud service model that provides networking capabilities (such as bandwidth, security, and connectivity) on-demand over the internet. It enables businesses to access network resources without investing in physical infrastructure.

ii) Explain two services that can be offered with Network-as-a-Service (NaaS) [4 marks]

Virtual Private Network (VPN) as a Service – Provides secure remote access to organizational resources over the internet.

Bandwidth on Demand (BoD) – Allows customers to scale network bandwidth dynamically based on their needs.

iii) Outline five benefits that can be acquired with Network-as-a-Service [5 marks]

Cost Savings – Reduces the need for expensive physical network infrastructure.

Scalability – Businesses can increase or decrease network capacity as required.

Flexibility – Supports remote access and global connectivity.

Security – Provides built-in encryption, firewalls, and access controls.

Simplified Management – Cloud providers handle network maintenance and upgrades.

c) Highlight three SaaS maturity models [3 marks]

Ad-Hoc (Level 1) – Basic SaaS applications with minimal scalability or multi-tenancy.

Configurable (Level 2) – Customizable applications with better resource sharing.

Scalable (Level 3) – Fully multi-tenant, scalable applications with high automation.

QUESTION 5

(a) You have been contacted as a cloud computing expert, to advise on steps to migrate

to the cloud. Explain SEVEN-STEP MODEL of migration into a cloud [7 marks]

Assess the Current IT Infrastructure – Identify existing workloads, applications, and infrastructure that need migration.

Select the Right Cloud Model – Choose between IaaS, PaaS, or SaaS, and decide on a public, private, hybrid, or community cloud.

Plan and Design the Migration – Develop a migration strategy, define objectives, and estimate costs.

Prepare the Cloud Environment – Set up the cloud infrastructure, security policies, and necessary integrations.

Migrate Applications and Data – Move applications, databases, and workloads using suitable migration tools.

Test and Optimize – Ensure migrated applications function correctly and optimize performance for cloud efficiency.

Manage and Monitor – Continuously monitor cloud performance, security, and scalability post-migration.

(b) A certain transport company wanted to incorporate cloud computing to assist them

render their services. Discuss the application areas of cloud computing in Transport

Company. [5 marks]

Fleet Management – Cloud-based GPS tracking enables real-time vehicle monitoring and route optimization.

Data Storage and Analytics – Stores vast amounts of transport data and uses analytics to improve operations.

Automated Ticketing Systems – Cloud-based booking and ticketing platforms allow easy access and management.

Traffic Management – Uses cloud AI to analyze traffic patterns and optimize routes for efficiency.

Customer Service – Chatbots and cloud CRM systems enhance customer interaction and complaint resolution.

(c) Explain the following cloud deployment methods: -

i) Public cloud [2 marks]

A cloud infrastructure that is owned and managed by third-party providers (e.g., AWS, Azure, Google Cloud).

Provides on-demand resources over the internet, suitable for startups and general businesses.

ii) Private cloud [2 marks]

A dedicated cloud infrastructure exclusively used by a single organization.

Offers higher security and customization, ideal for banks and government institutions.

iii) Community cloud [2 marks]

A shared cloud infrastructure used by organizations with similar needs (e.g., healthcare, education).

Provides collaborative data sharing and security compliance for specific industries.

iv) Hybrid cloud [2 marks]

Combines public and private cloud services to balance cost, security, and performance.

Allows critical applications to run on a private cloud while less sensitive data uses the public cloud.